UNIT V

Deep Learning and Expert System

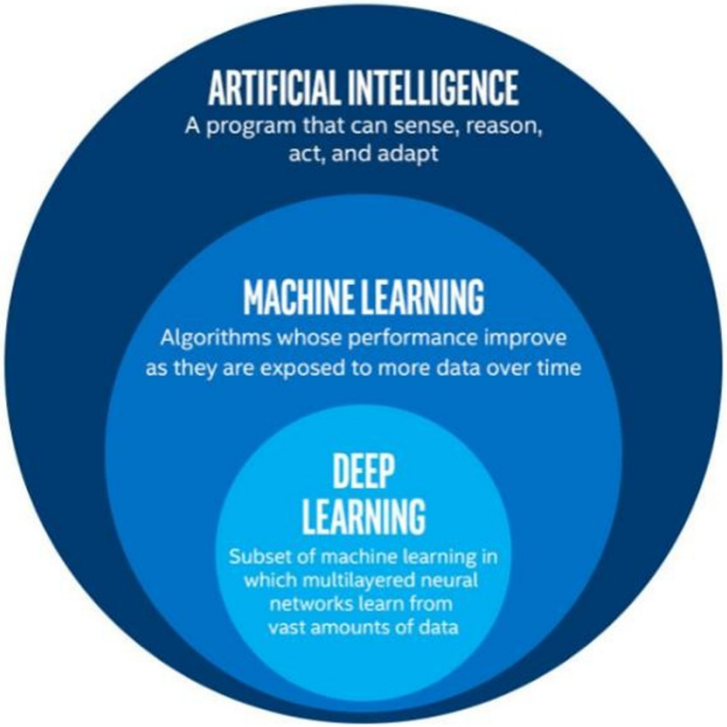
Outline

Deep Learning :

- Neurons, Linear Perceptrons as Neurons, Neural Nets Architecture/ Design, Working of Neural Nets, Layers of Neural Networks and Deep learning, Activation Functions,
- Feed Forward Neural Networks, Limitations of Neurons Deep Belief Networks (DBNs) Large Scale DBNs, Large Scale Convolutional Neural Networks,
- Deep Learning for Big Data, Deep Learning from High Volumes of Data, Deep Learning from High Variety of Data ,Deep Learning for High Velocity of Data , Local Minima in Deep Networks, Rearranging Neurons in a layer of a Neural Network, Spurious Local Minima in Deep Networks.
- **Expert System**:
- Characteristics, Components, Development, Knowledge Engineering, Application.

Machine Learning and Deep Learning

- Machine Learning and Deep Learning are the two main concepts of Data Science and the subsets of Artificial Intelligence.
- Most of the people think the machine learning, deep learning, and as well as artificial intelligence as the same buzzwords.
- But in reality, all these terms are different but related to each other.



Machine Learning

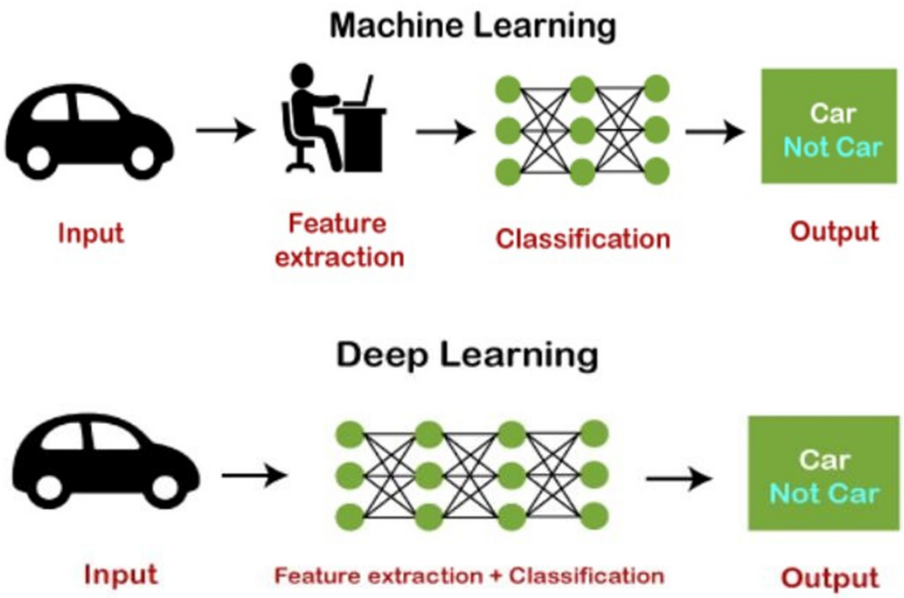
- Machine learning is a part of artificial intelligence and growing technology that enables machines to learn from past data and perform a given task automatically.
- Machine Leaning allows the computers to learn from the experiences by its own, use statistical methods to improve the performance and predict the output without being explicitly programmed.
- The popular applications of ML are Email spam filtering, product recommendations, online fraud detection, etc.
- Some useful ML algorithms are:
  - **Decision Tree algorithm**
  - **Naïve Bayes**
  - **Random Forest**
  - **K-means clustering**
  - **KNN algorithm**

Deep Learning

- *Deep Learning is the subset of machine learning or can be said as a special kind of machine learning*.
- It works technically in the same way as machine learning does, but with different capabilities and approaches.
- It is inspired by the functionality of human brain cells, which are called neurons, and leads to the concept of artificial neural networks. It is also called a **deep neuralnetwork** or **deep neural learning**.
- In deep learning, models use different layers to learn and discover insights from the data.
- Some popular applications of deep learning are self-driving cars, language translation, natural language processing, etc.
- Some popular deep learning models are:
  - Classic Neural Networks
  - Deep Belief Networks
  - Convolutional Neural Network
  - Recurrent Neural Network

Machine Learning and Deep Learning

- **Example: To identify the image of a car.**
- For this, the **Machine Learning model** takes images of car as input, extracts the different features of images such as shape, height, color etc., applies the classification algorithm, and predict the output.
- The **Deep Learning model** takes the images as the input and feed it directly to the algorithms without requiring any manual feature extraction step. The images pass to the different layers of the artificial neural network and predict the final output.

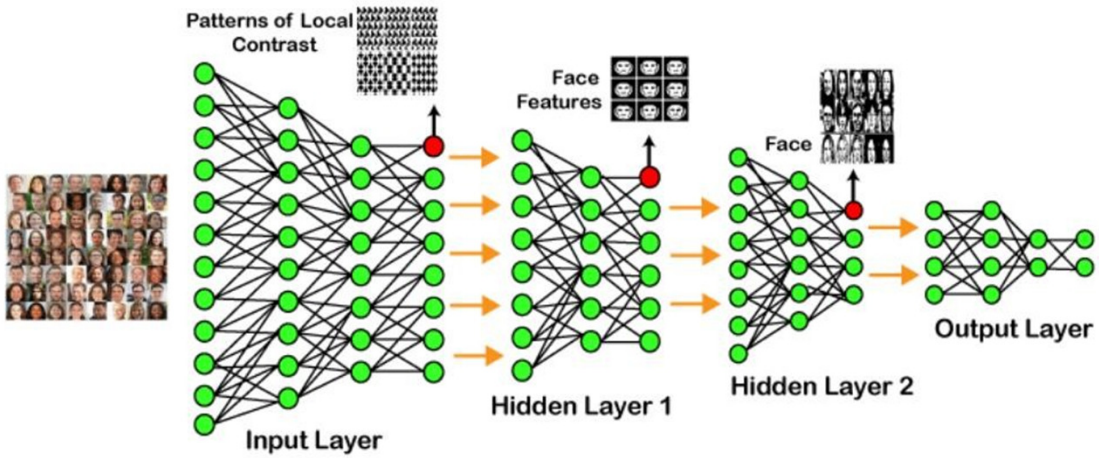


Deep Learning

- Deep learning is based on the branch of machine learning, which is a subset of artificial intelligence. Since neural networks imitate the human brain and so deep learning will do.
- In deep learning, nothing is programmed explicitly. Basically, it is a machine learning class that makes use of numerous nonlinear processing units so as to perform feature extraction as well as transformation. The output from each preceding layer is taken as input by each one of the successive layers.
- Deep learning models are capable enough to focus on the accurate features themselves by requiring a little guidance from the programmer and are very helpful in solving out the problem of dimensionality. Deep learning algorithms are used, especially when we have a huge no of inputs and outputs.
- Deep learning is implemented with the help of Neural Networks, and the idea behind the motivation of Neural Networks is the biological neurons, which is nothing but a brain cell.
- *Deep learning is a collection of statistical techniques of machine learning for learning feature hierarchies that are actually based on artificial neural networks*.

Example of Deep Learning

- In the given example, we provide the raw data of images to the first layer of the input layer.
- After then, these input layer will determine the patterns of local contrast that means it will differentiate on the basis of colors, luminosity, etc.
- Then the 1st hidden layer will determine the face feature, i.e., it will fixate on eyes, nose, and lips, etc.



- And then, it will fixate those face features on the correct face template. So, in the 2nd hidden layer, it will actually determine the correct face here as it can be seen in the above image, after which it will be sent to the output layer.
- Likewise, more hidden layers can be added to solve more complex problems, for example, if you want to find out a particular kind of face having large or light complexions. So, as and when the hidden layers increase, we are able to solve complex problems.

Deep Learning Applications

- ***Self-Driving Cars***
- In self-driven cars, it is able to capture the images around it by processing a huge amount of data, and then it will decide which actions should be incorporated to take a left or right or should it stop. So, accordingly, it will decide what actions it should take, which will further reduce the accidents that happen every year.
- ***Voice Controlled Assistance***
- In voice control assistance, the **Siri** (a built-in, voice-controlled personal assistant available for Apple users) is the one thing that comes into our mind. So, you can tell Siri whatever you want it to do it for you, and it will search it for you and display it for you.
- ***Automatic Image Caption Generation***
- Whatever image that you upload, the algorithm will work in such a way that it will generate caption accordingly. If you say blue colored eye, it will display a blue-colored eye with a caption at the bottomof the image.
- ***Automatic Machine Translation***
- With the help of automatic machine translation, we are able to convert one language into another with the help of deep learning.

Deep Learning

- **Limitations**
  - It only learns through the observations.
  - It comprises of biases issues.
- **Advantages**
  - It lessens the need for feature engineering.
  - It eliminates all those costs that are needless.
  - It easily identifies difficult defects.
  - It results in the best-in-class performance on problems.

Disadvantages

- It requires an ample amount of data.
- It is quite expensive to train.
- It does not have strong theoretical groundwork.

Artificial Neural Networks

- At earlier times, the conventional computers incorporated algorithmic approach that is the computer used to follow a set of instructions to solve a problem unless those specific steps need that the computer need to follow are known the computer cannot solve a problem.
- So, obviously, a person is needed in order to solve the problems or someone who can provide instructions to the computer so as to how to solve that particular problem.
- It actually restricted the problem-solving capacity of conventional computers to problems that we already understand and know how to solve. But what about those problems whose answers are not clear, so that is where our traditional approach face failure and so Neural Networks came into existence.
- Neural Networks processes information in a similar way the human brain does, and these networks actually learn from examples, you cannot program them to perform a specific task. They will learn only from past experiences as well as examples, which is why you don't need to provide all the information regarding any specific task. So, that was the main reason why neural networks came into existence.

Artificial Neural Networks

- The term "Artificial neural network" refers to a biologically inspired sub-field of artificial intelligence modeled after the brain.
- An Artificial neural network is usually a computational network based on biological neural networks that construct the structure of the human brain.
- Similar to a human brain has neurons interconnected to each other, artificial neural networks also have neurons that are linked to each other in various layers of the networks.
- These neurons are known as nodes.
- A biological neuron receives input from other sources, combines them in some way, followed by performing a nonlinear operation on the result, and the output is the final result.

Artificial Neural Networks

- The **dendrites** will act as a receiver that receives signals from other neurons, which are then passed on to the cell body.
- The **cell body** will perform some operations that can be a summation, multiplication, etc.
- After the operations are performed on the set of input, then they are transferred to the next neuron via **axon**, which is the transmitter of the signal for the neuron.

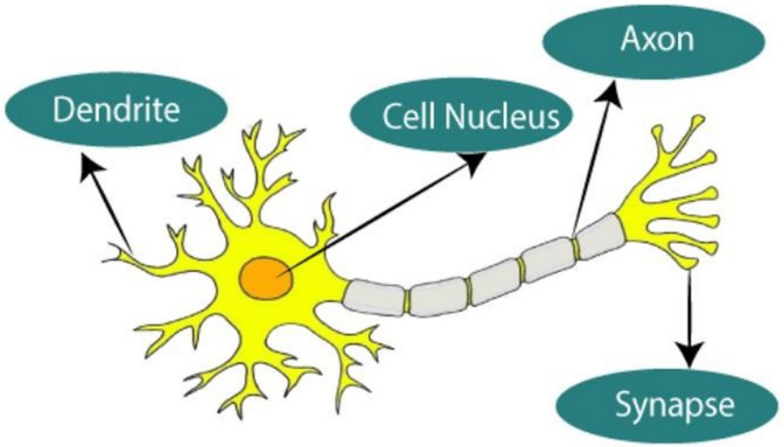


Figure: The typical diagram of Biological Neural Network.

The Architecture of Artificial Neural Network

- In order to define a neural network that consists of a large number of artificial neurons, which are termed units arranged in a sequence of layers.
- Artificial Neural Network primarily consists of three layers:
- **Input Layer:** As the name suggests, it accepts inputs in several different formats provided by the programmer.
- **Hidden Layer:** The hidden layer presents in-between input and output layers. It performs all the calculations to find hidden features and patterns.
- **Output Layer:** The input goes through a series of transformations using the hidden layer, which finally results in output that is conveyed using this layer.

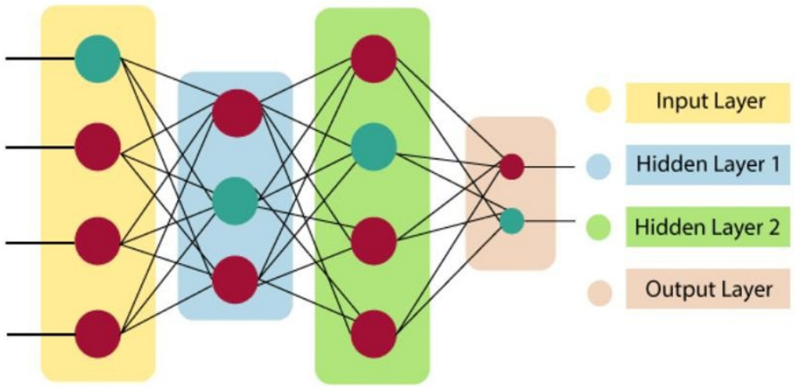


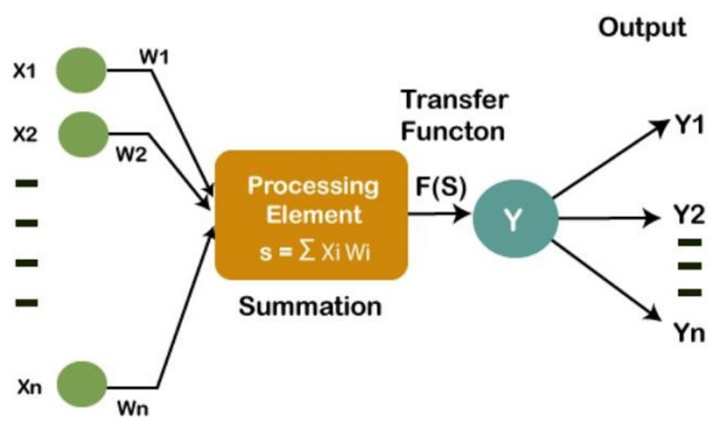
Figure: Basic Architecture of Artificial Neural Network

The Architecture of Artificial Neural Network

- The artificial neural network takes input and computes the weighted sum of the inputs and includes a bias. This computation is represented in the form of a transfer function.
- It determines weighted total is passed as an input to an activation function to produce the output.
- Activation functions choose whether a node should fire or not.
- Only those who are fired make it to the output layer.
- There are distinctive activation functions available that can be applied upon the sort of task we are performing.

Working of Artificial Neural Network

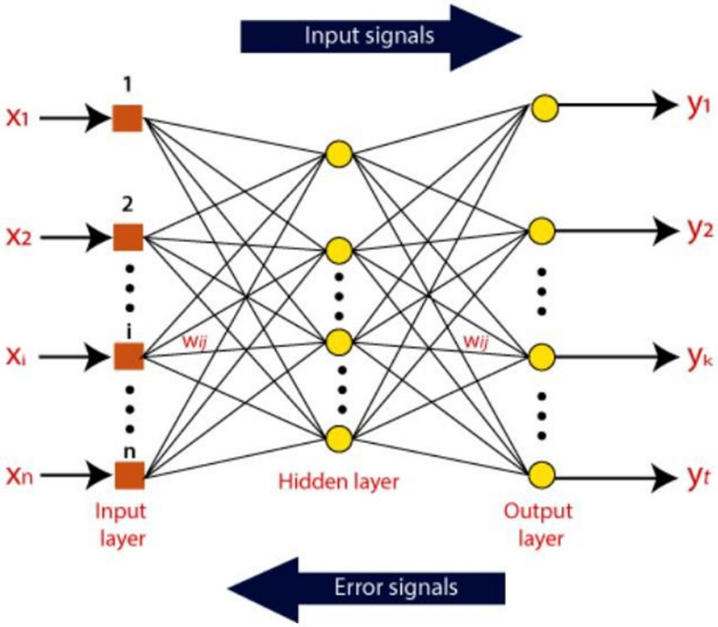
- Artificial Neural Network can be best represented as a weighted directed graph, where the artificial neurons form the nodes.
- The association between the neurons outputs and neuron inputs can be viewed as the directed edges with weights.
- The Artificial Neural Network receives the input signal from the external source in the form of a pattern and image in the form of a vector. These inputs are then mathematically assigned by the notations x(n) for every n number of inputs.



- The inputs (x) are fed into the input layer, which undergoes multiplication with the allotted weights (w) followed by experiencing addition in order to form weighted sums.
- Then these inputs weighted sums with their corresponding weights are executed on the related activation function or transfer function.

Working of Artificial Neural Network

- In general terms, these weights normally represent the strength of the interconnection between neurons inside the artificial neural network. All the weighted inputs are summarized inside the computing unit.
- If the weighted sum is equal to zero, then bias (helps in the adjustment of the curve of activation function so as to accomplish a precise output) is added to make the output non-zero or something else to scale up to the system's response.
- Bias has the same input, and weight equals to 1. Here the total of weighted inputs can be in the range of 0 to positive infinity.
- Here, to keep the response in the limits of the desired value, a certain maximum value is benchmarked, and the total of weighted inputs is passed through the activation function.



Activation function

- The activation function refers to the set of transfer functions used to achieve the desired output.
- The main objective of the activation function is to perform a mapping of a weighted sum upon the output.
- The transformation function comprises of activation functions such as tanh, sigmoid, etc.
- The activation function is categorized into two main parts:
  1. Linear Activation Function
  2. Non-Linear Activation Function

Artificial Neural Network and Deep Learning

- **Artificial Neural Network** or **Neural Network** was modeled after the human brain. Human has a mind to think and to perform the task in a particular situation, but how can a machine do that? For this purpose, an artificial brain was designed, which is known as a Neural Network. As the human brain has neurons for passing information, similarly neural network has nodes to perform that task. Nodes are the mathematical functions.
- A Neural Network is based on the structure and functions of biological Neural Networks. A Neural Network itself changes or learn based on input and output. The information that flows through the network affect the structure of the artificial Neural Network because of its learning and changing property.
- **Deep Learning Neural Network** is an advanced form of neural network. Unlike simple Neural Network, Deep Learning Neural Network have more than one hidden layer. Deep Learning Neural Network gets the more complex dataset as that your model is able to learn from.

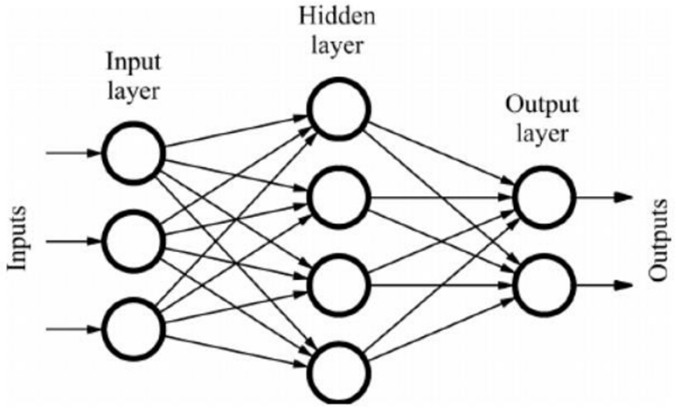
Artificial Neural Network and Deep Learning Neural Network

Types of Deep Learning

- Deep learning follows a distributed approach to managing big data. The method assumes that the data gets generated considering numerous factors, different time, and various levels. Deep learning facilitates the arrangement and processing of the data into different layers according to its time (occurrence), its scale, or nature. Deep learning is often associated with artificial neural networks.
- **There are different deep learning models:**
  - Feed forward Neural Network
  - Deep Belief Networks
  - Convolutional Neural Network
  - Radial Basis Function Neural Network
  - Recurrent Neural Network
  - Modular Neural Network

Feed Forward Neural Networks

- The **feed forward model** is the simplest form of neural network as information is only processed in one direction.
- While the data may pass through multiple hidden nodes, it always moves in one direction and never backwards.
- A Feed Forward Neural Network is commonly seen in its simplest form as a *single layer perceptron*.



- In this model, a series of inputs enter the layer and are multiplied by the weights. Each value is then added together to get a sum of the weighted input values.
- If the sum of the values is above a specific threshold, usually set at zero, the value produced is often 1, whereas if the sum falls below the threshold, the output value is -1.

Feed Forward Neural Networks

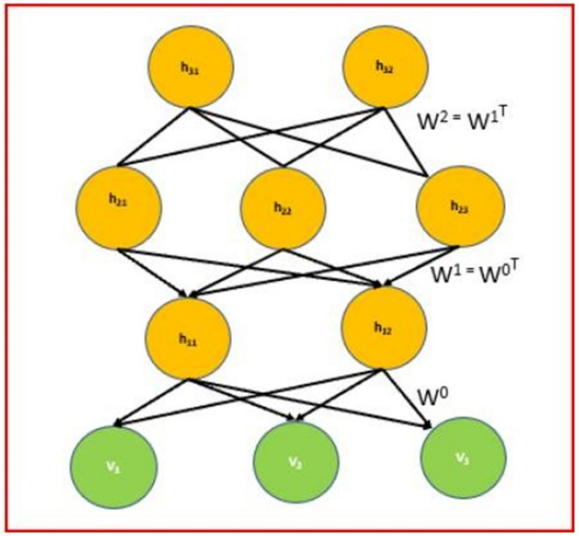
- The ***single layer perceptron*** is an important model of feed forward neural networks and is often used in classification tasks. Furthermore, single layer perceptrons can incorporate aspects of machine learning.
- In ***multi-layered perceptrons***, the process of updating weights is nearly analogous, however the process is defined more specifically as back-propagation. In such cases, each hidden layer within the network is adjusted according to the output values produced by the final layer.
- There are many applications of Feedforward neural networks, such as **speech recognition** and **computer vision**. It is easier to maintain these types of Neural Networks and also has excellent responsiveness to noisy data.

Deep Belief Networks (DBNs)

- Deep belief networks are algorithms that use probabilities and unsupervised learning to produce outputs.
- They are composed of binary latent variables, and they contain both undirected layers and directed layers.
- Unlike other models, each layer in deep belief networks learns the entire input.
- In convolutional neural networks, the first layers only filter inputs for basic features, such as edges, and the later layers recombine all the simple patterns found by the previous layers.
- Deep belief networks, on the other hand, work globally and regulate each layer in order.
- In Deep belief networks there are many applications such as--Image recognition, Video recognition, Motion-capture data, etc.

Deep Belief Networks (DBNs)

- The network is like a stack of Restricted Boltzmann Machines (RBMs), where the nodes in each layer are connected to all the nodes in the previous and subsequent layer.
- However, unlike RBMs, nodes in a deep belief network do not communicate laterally within their layer.
- A network of symmetrical weights connect different layers.
- The connections in the top layers are undirected and associative memory is formed from the connections between them. The connections in the lower levels are directed.
- The nodes in the hidden layer fulfill two role -- they act as a hidden layer to nodes that precede it and as visible layers to nodes that succeed it.
- These nodes identify the correlations in the data.

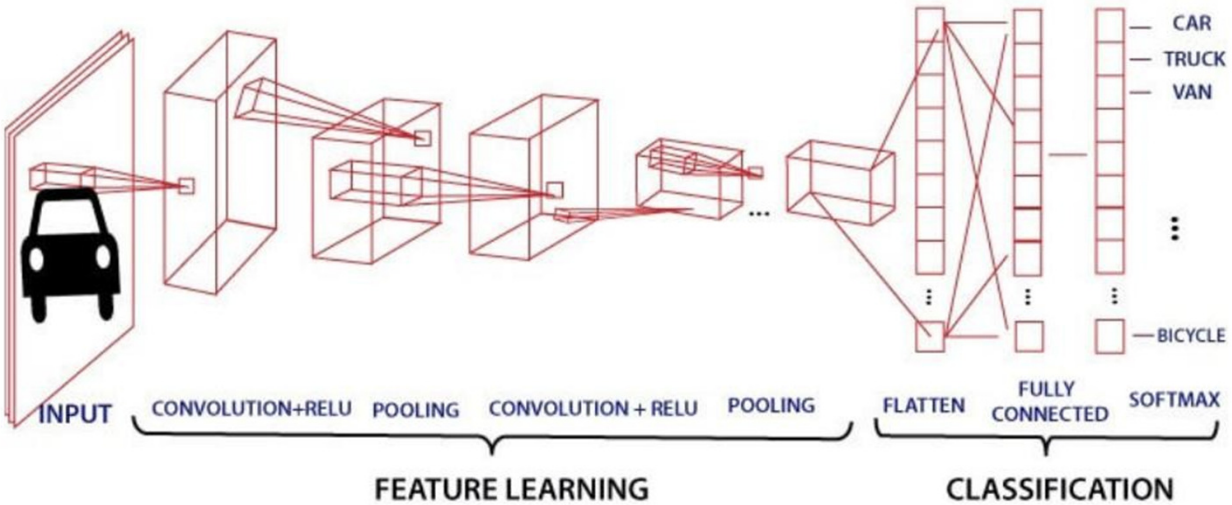


Convolution Neural Networks

- **Convolutional Neural Network** is one of the main categories to do image classification and image recognition in neural networks.
- Scene labeling, objects detections, and face recognition, etc., are some of the areas where convolutional neural networks are widely used.
- CNN takes an image as input, which is classified and process under a certain category such as dog, cat, lion, tiger, etc.
- The computer sees an image as an array of pixels and depends on the resolution of the image. Based on image resolution, it will see as **h * w * d**, where h= height w= width and d= dimension.
- For example, An RGB image is **6 * 6 * 3** array of the matrix, and the grayscale image is **4 * 4 * 1** array of the matrix.

Convolution Neural Networks

- In CNN, each input image will pass through a sequence of convolution layers along with pooling, fully connected layers, filters (Also known as kernels).
- After that, we will apply the Soft-max function to classify an object with probabilistic values 0 and 1.



Convolution Neural Networks

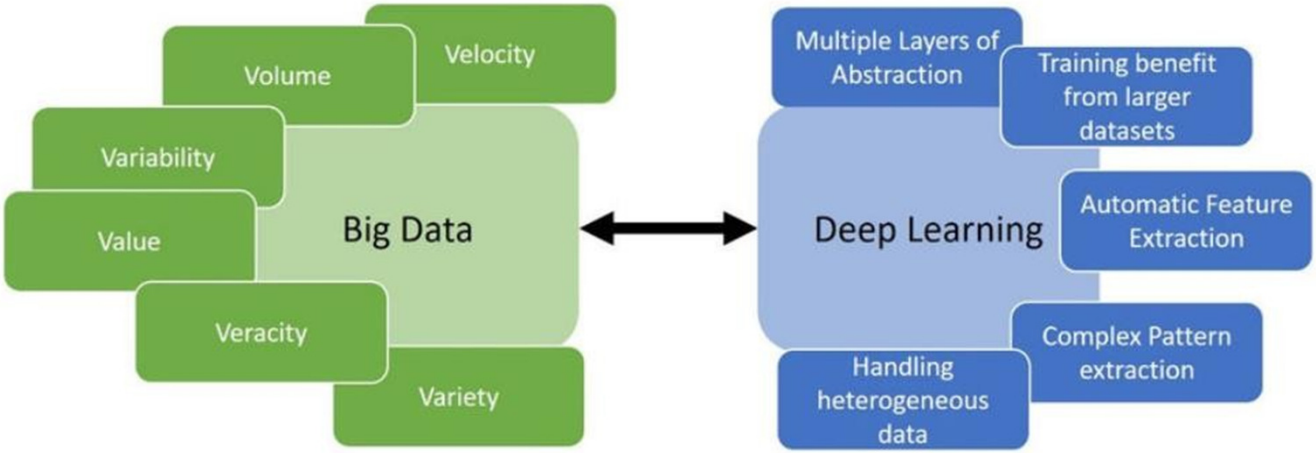
- The basic layers of CNN are:
- **Convolution layer** : a “filter”, sometimes called a “kernel”, is passed over the image, viewing a few pixels at a time (for example, 3X3 or 5X5).
- The convolution operation is a dot product of the original pixel values with weights defined in the filter. The results are summed up into one number that represents all the pixels the filter observed.
- **Activation layer**: the convolution layer generates a matrix that is much smaller in size than the original image.
- This matrix is run through an activation layer, which introduces non-linearity to allow the network to train itself via backpropagation. The activation function is typically ReLu.

Convolution Neural Networks

- **Pooling layer**: “pooling” is the process of further down sampling and reducing the size of the matrix.
- A filter is passed over the results of the previous layer and selects one number out of each group of values (typically the maximum, this is called max pooling).
- This allows the network to train much faster, focusing on the most important information in each feature of the image.
- **Fully connected layer**: a traditional multilayer perceptron structure. Its input is a one-dimensional vector representing the output of the previous layers.
- Its output is a list of probabilities for different possible labels attached to the image (e.g. dog, cat, bird). The label that receives the highest probability is the classification decision.
- There may be multiple activation and pooling layers, depending on the CNN architecture.

Deep Learning for Big Data

- The term big data refers to the rapid increase in data in the organizations and societies of today.
- Deep learning is a recent and powerful tool, which organizations and cities use to obtain value from big data.



Deep Learning for Big Data

- The term big data refers to the proliferation of data in modern society and business, along three dimensions:
- **Volume**—growing quantities of data, both structured and unstructured. Unstructured data is the fastest growing, and can take the shape of social media feeds, web analytics data, sensor data or documents generated by humans in an organization.
- **Velocity**—data is flowing at an increasing rate. There is a need to ingest, store and process massive data streams, either in real time as they are received, or in retrospect.
- **Variety**—traditional data types were structured and controllable by organizations. Today the number of data types is rapidly growing, from structured to semi structured and unstructured data, including text, audio, video from millions of possible sources.

Deep Learning for Big Data

- Big data is the fuel of the modern economy. It is used by the world's biggest companies to provide products and services that are changing the face of society.
- The digital economy, digital lifestyle, mobile devices and applications that have become an inseparable part of daily life, are all driven by big data. But the raw data, like crude oil, isn't useful on its own.
- Over the past two decades, the industry has developed technology that can ingest, store, process, analyze and derive value quickly from big datasets.
- But some datasets are a harder nut to crack—complex, unstructured data is difficult to comprehend using traditional analytics techniques.
- This is where deep learning comes in—it can help make sense of big data often through the use of a deep learning platform.

Applications of Deep Learning for Big Data

- Deep learning, with artificial neural networks at its core, is a new and powerful technique that can be used to derive value from big data.
- Most of the data today is unstructured, and deep learning algorithms are very effective at learning from, and generating predictions for, wildly unstructured data.
- Following are several ways deep learning is being applied to big data sets.
  - Text Classification and Automatic Tagging
  - Automatic Image Caption Generation
  - Deep Learning in Finance
  - Deep Learning in Healthcare

Applications of Deep Learning for Big Data

1. Text Classification and Automatic Tagging

- Deep learning architectures including Recurrent Neural Networks and Convolutional Neural Networks are used to process free text, perform sentiment analysis and identify which categories or types it belongs to.
- This can help search through, organize and make use of huge unstructured datasets.

2. Automatic Image Caption Generation

- Deep learning is used to identify the contents of an image and automatically generate descriptive text, turning images from unstructured to structured and searchable content.
- This involves the use of Convolutional Neural Networks, in particular very large networks like ResNet, to perform object detection, and then using Recurrent Neural networks to write coherent sentences based on object classification.

Applications of Deep Learning for Big Data

- **3. Deep Learning in Finance**

- Today, most financial transactions are electronic. Stock markets generate huge volumes of data reflecting buy and sell actions, and the resulting financial metrics such as stock prices. Deep learning can ingest these huge data volumes, understand the current market position and create an accurate model of the probabilities of future price movements.
- However deep learning is mainly used for analyzing macro trends or making one-time decisions such as analyzing the possibility of company bankruptcy; it is still limited in its ability to drive real-time buying decisions.

- **4. Deep Learning in Healthcare**

- Deep learning, particularly computer vision algorithms, are used to help diagnose and treat patients.
- Deep learning algorithms analyze blood samples, track glucose levels in diabetic patients, detect heart problems, analyze images to detect tumors, can diagnose cancer, and are able to detect osteoarthritis from an MRI scan before damage is caused to bone structures.

Expert System

- An expert system is a computer program that is designed to solve complex problems and to provide decision-making ability like a human expert.
- It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.
- The expert system is a part of AI, and the first ES was developed in the year 1970, which was the first successful approach of artificial intelligence. It solves the most complex issue as an expert by extracting the knowledge stored in its knowledge base.
- The system helps in decision making for complex problems using **both facts and heuristics like a human expert**.
- It is called so because it contains the expert knowledge of a specific domain and can solve any complex problem of that particular domain.
- These systems are designed for a specific domain, such as **medicine, science,** etc.

Expert System

- The performance of an expert system is based on the expert's knowledge stored in its knowledge base.
- The more knowledge stored in the KB, the more that system improves its performance.
- One of the common examples of an ES is a suggestion of spelling errors while typing in the Google search box.

- It is important to remember that an expert system is not used to replace the human experts; instead, it is used to assist the human in making a complex decision.
- These systems do not have human capabilities of thinking and work on the basis of the knowledge base of the particular domain.

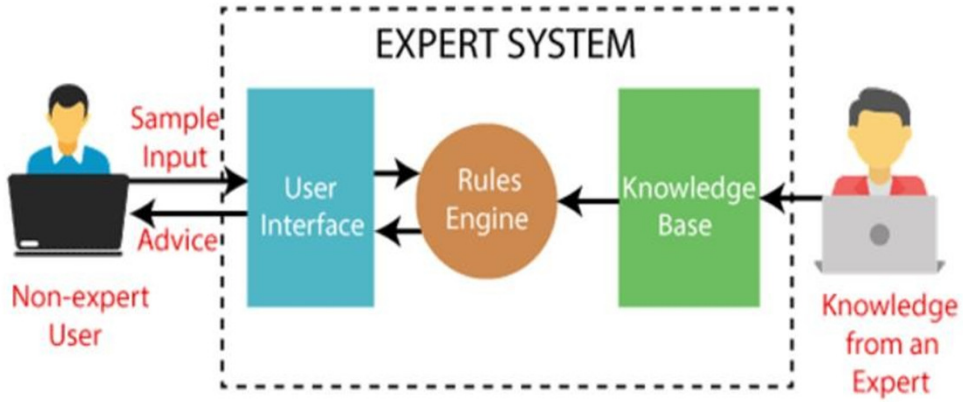


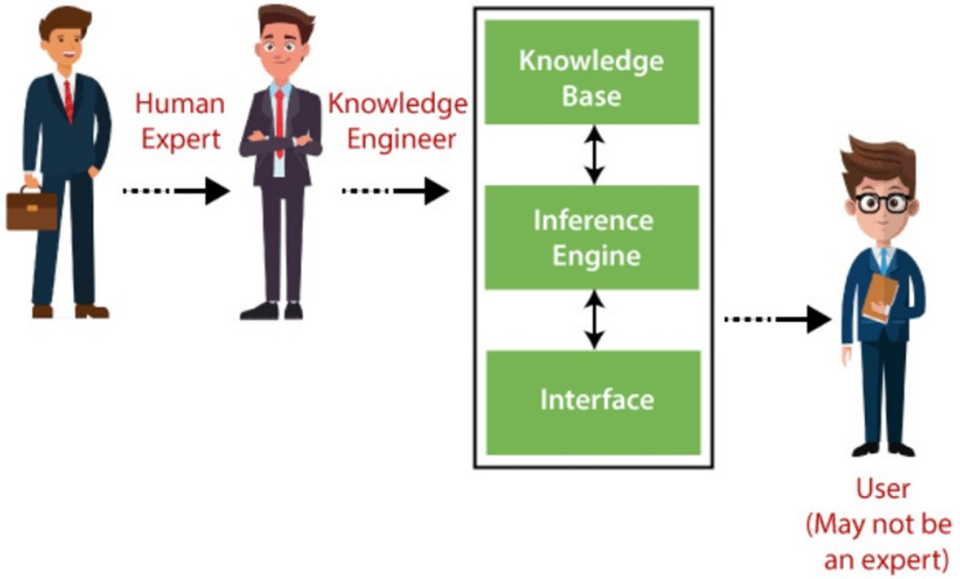
Figure: Block diagram of working of an expert system

Characteristics of Expert System

- **High Performance:** The expert system provides high performance for solving any type of complex problem of a specific domain with high efficiency and accuracy.
- **Understandable:** It responds in a way that can be easily understandable by the user. It can take input in human language and provides the output in the same way.
- **Reliable:** It is much reliable for generating an efficient and accurate output.
- **Highly responsive:** ES provides the result for any complex query within a very short period of time.

Components of Expert System

- An expert system mainly consists of three components:
- **User Interface**
- **Inference Engine**
- **Knowledge Base**



Components of Expert System

1. User Interface

- With the help of a user interface, the expert system interacts with the user, takes queries as an input in a readable format, and passes it to the inference engine. After getting the response from the inference engine, it displays the output to the user. In other words, **it is an interface that helps a non-expert user to communicate with the expert system to find a solution**.

2. Inference Engine(Rules of Engine)

- The inference engine is known as the brain of the expert system as it is the main processing unit of the system. It applies inference rules to the knowledge base to derive a conclusion or deduce new information. It helps in deriving an error-free solution of queries asked by the user.
- With the help of an inference engine, the system extracts the knowledge from the knowledge base.

Components of Expert System

- There are two types of inference engine:
  - **Deterministic Inference engine:** The conclusions drawn from this type of inference engine are assumed to be true. It is based on **facts** and **rules**.
  - **Probabilistic Inference engine:** This type of inference engine contains uncertainty in conclusions, and based on the probability.
- Inference engine uses the below modes to derive the solutions:
  - **Forward Chaining:** It starts from the known facts and rules, and applies the inference rules to add their conclusion to the known facts.
  - **Backward Chaining:** It is a backward reasoning method that starts from the goal and works backward to prove the known facts.

Components of Expert System

3. Knowledge Base

- The knowledgebase is a type of storage that stores knowledge acquired from the different experts of the particular domain.
- It is considered as big storage of knowledge. The more the knowledge base, the more precise will be the Expert System.
- It is similar to a database that contains information and rules of a particular domain or subject.
- One can also view the knowledge base as collections of objects and their attributes.
- Such as a Lion is an object and its attributes are it is a mammal, it is not a domestic animal, etc.

Development of Expert System

- Working of an expert system by taking an example of MYCIN ES:
  1. Firstly, ES should be fed with expert knowledge. In the case of MYCIN, human experts specialized in the medical field of bacterial infection, provide information about the causes, symptoms, and other knowledge in that domain.
  2. The KB of the MYCIN is updated successfully. In order to test it, the doctor provides a new problem to it. The problem is to identify the presence of the bacteria by inputting the details of a patient, including the symptoms, current condition, and medical history.
  3. The ES will need a form to be filled by the patient to know the general information about the patient, such as gender, age, etc.

Development of Expert System

4.Now the system has collected all the information, so it will find the solution for the problem by applying if-then rules using the inference engine and using the facts stored within the KB.

5. In the end, it will provide a response to the patient by using the user interface.

- **There are three primary participants in the building of Expert System:**

1. **Expert:** The success of an ES much depends on the knowledge provided by human experts. These experts are those persons who are specialized in that specific domain.
2. **Knowledge Engineer:** Knowledge engineer is the person who gathers the knowledge from the domain experts and then codifies that knowledge to the system according to the formalism.
3. **End-User:** This is a particular person or a group of people who may not be experts, and working on the expert system needs the solution or advice for his queries, which are complex.

Applications of Expert System

- **In designing and manufacturing domain**

  It can be broadly used for designing and manufacturing physical devices such as camera lenses and automobiles.

- **In the knowledge domain**

  These systems are primarily used for publishing the relevant knowledge to the users. The two popular ES used for this domain is an advisor and a tax advisor.

- **In the finance domain**

  In the finance industries, it is used to detect any type of possible fraud, suspicious activity, and advise bankers that if they should provide loans for business or not.

Applications of Expert System

- **In the diagnosis and troubleshooting of devices**
  In medical diagnosis, the ES system is used, and it was the first area where these systems were used.

- **Planning and Scheduling**
  The expert systems can also be used for planning and scheduling some particular tasks for achieving the goal of that task.

END
of
UNIT V